

Assignment 03

(Handwritten in A4 pages, like assignment 01)

1. x and n are input through keyboard. Write a program to compute x^n , $n!$, nC_r , nP_r
2. Write a program to determine the **GCD** (greatest common divisor) and **LCM** (least common multiple) of 3 numbers.
3. Find out the sum of each of the following series. n is the input from user for series
 - (i) $3 + 11 + 19 + \dots + 1691$.
 - (ii) $7 + 20 + 33 + \dots$ (up to 100 th term)
 - (iii) $5 - 11 + 17 - \dots$ (up to 75 th term)
 - (iv) $1 + (1 + 2) + (1 + 2 + 3) + \dots + (1 + 2 + 3 + \dots + n)$
 - (v) $1 + \frac{2^2}{2!} + \frac{3^2}{3!} + \dots + \frac{n^2}{n!}$
 - (vi) $2 * 7 * 12 * \dots * 37$
4. Write a program to determine all **prime numbers** within the range $[a \dots b]$ where a & b are input through keyboard.
5. Construct the following table. Here n is input from the user.

1	2	3	...	n
2	4	6	...	$2n$
3	6	9	...	$3n$
.	.	.	...	.
.	.	.	...	.
.	.	.	...	.
n	$2n$	$3n$	...	nn
6. Write a program to find out first n **perfect number** where n is the input from user.
7. Write a program to find first n **Fibonacci number** where n is the input from user.
8. Write a program to show the following triangle/rectangle of '*'s or numbers. Take n as input from user to determine the number of rows of the structure. (eg: $n = 5$)

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    * * *
  * * * * *
* * * * * * *
* * * * * * * *
```

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      *
    *  *
  *    *
*      *
* * * * * *
```

```
      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```

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1 2 3 2 1
1 1
1 1
1 1
1 2 3 2 1

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 1
 1 2 1
 1 2 3 2 1
 1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
 1 2 3 4 3 2 1
 1 2 3 2 1
 1 2 1
 1

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```

1 2 3 4 5 4 3 2 1
 1 2 3 4 3 2 1
   1 2 3 2 1
    1 2 1
     1
    1 2 1
   1 2 3 2 1
  1 2 3 4 3 2 1
 1 2 3 4 5 4 3 2 1

```

9. Write a program to print out all **Armstrong numbers** between 1 and 10000. If sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. For example, $153 = (1*1*1) + (5*5*5) + (3*3*3)$.
10. Write a program to calculate how many 5 digit numbers can be created if the following terms apply :
 - (i) the leftmost digit is even
 - (ii) the second digit is odd
 - (iii) the third digit is a non even prime
 - (iv) the fourth and fifth are two random digits not used before in the number.